

K. M. Abesh Ahsan

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Education

[Islamic University of Technology \(IUT\)](#)

Bachelor of Science in Computer Science and Engineering (CSE)

CGPA: 3.72/4.0

Jun 2021 – October 2025

Gazipur, Bangladesh

Experience

[Battery Low Interactive](#) – Game Development Trainee

Oct 2024

During my industrial training as a Game Developer Trainee, I learned to conceptualize and program games from inception using Unity and C#. I engineered several mini-games to grasp mechanics, scoring algorithms, and physics-driven interactions. For my final project, I developed a 2D centrifugal-force game in which players maneuvered a stone to strike targets, implementing custom game logic, score tracking, and responsive UI components. I also integrated animations, curated assets, and designed scenes. I also explored the fundamentals of AR/VR during that time.

Research

[Weakly Supervised Semantic Segmentation With Image Labels](#)

January 2024 – October 2025

I designed a transformer-based approach to enhance object detail recognition in weakly supervised segmentation. Using UniCL with a Swin Transformer, I improved class activation map (CAM) quality and fine-grained feature extraction via windowed attention. My method incorporated advanced CAM refinement, affinity calculations, and encoder intermediate features. This model generated highly detailed masks for most classes, though accuracy dropped for a few categories like person and chair. Achieving a mean IoU of 50%, the approach delivered substantial fine-grained segmentation improvements, despite being below the state-of-the-art result of 74%.

Skills

Programming Languages: C, C++, Python, Java, C#, Dart, JavaScript, TypeScript, HTML, CSS, Bash, SQL**Frameworks and Libraries:** Flutter, React, Next.js, Node.js, Express.js, JavaFX, PyQt6, Bootstrap, Tailwind**Tools and Technologies:** Git, Docker, Visual Studio, IntelliJ, Unity, MySQL, MongoDB, Realm DB, Pusher, Upstash Redis

Projects

[Inner Voices](#)

2026 — Flutter, BLoC, Realm DB, GDrive API, Biometric, E2E

A secure, private digital journal app that provides a safe space for personal thoughts and reflections. It prioritizes privacy through local PIN and biometric app locking. The app features rich text editing, offline-first storage for writing and reading entries without internet, and end-to-end encrypted cloud backups to the user's Google Drive. It includes smart sync monitoring to show local vs. cloud status, along with dynamic theming (light, dark, system) for a smooth, intuitive experience across devices.

[FunGames](#) (In progress)

2026 — Next.js, TypeScript, Tailwind, Zustand, Pusher, Upstash Redis

Modern browser-based gaming platform featuring an expanding suite of mini-games (Memory Match live; Tic-Tac-Toe, Snake, Chess + more in active development). Offers anonymous no-login entry, real-time multiplayer via WebSockets, session-persistent user IDs and usernames, responsive cross-device design, optimized low-latency performance with serverless Redis storage, and smooth instant-play gameplay for casual, high-engagement sessions.

[VS-Leet](#)

2025 — VS Code Extension API, Node.js, React.js, Tailwind CSS

A VS Code extension that eliminates browser-editor context switching for LeetCode users, letting you browse/filter problems, code in C++/Java/Python/JS/TS, run tests with real-time feedback, and submit solutions—all within your IDE to streamline DSA practice and crush technical interviews.

[Photo Wizard](#)

2024 — NumPy, PyQt6

A desktop app with image editing features including crop, resize, blur, sharpen, exposure, brightness, contrast, saturation, warmth, mirror, rotate, and filters. Supports custom filter creation. All processing implemented **from scratch** using NumPy, without OpenCV or PIL.

[Tilawah](#)

2024 — jQuery, JavaScript, Axios, EJS, Node.js, Express.js, MySQL

A single-page web application (SPA) for listening to Quran recitations from various Qaris. Features include seamless playback with resume support, search by Surah or Qari, favorites, playlist management, and live updates – all implemented using EJS and jQuery without modern frontend frameworks.

Programming Achievements

[IUT Intra First-Year Programming Contest 2021](#)

2021 — 1st out of 32 teams

[CodeRush 1.0: Programming Competition](#)

2023 — 7th out of 37 teams